Quizzes - Topology 0*

December 10, 2025

1 Session 03/12/2025

These quizzes are completely optional; it is just an additional opportunity to practice and get familiar with the material from the lecture.

1. Can two metrics on $\mathbb{R}$ give rise to two different topologies? Three metrics?
2. Find a set with infinitely many non homeomorphic topologies?
3. Let (X, τ) be a topological space. If $A \subseteq X$ contains its boundary, is A closed?
4. Give examples of a topological space (X, τ) such that
   - X is homeomorphic to a proper subspace $A \subsetneq X$ with the subspace topology.
   - X is *not* homeomorphic to *any* proper subspace $A \subsetneq X$ with the subspace topology.
5. In a Hausdorff space, can we separate points by closed sets?
6. If A, B are connected subspaces of (X, τ), and $A \subseteq C \subseteq B$, is C connected?
7. Let $\tau_1 \subseteq \tau_2$ be two different topologies on X. If (X, τ_1) is connected, resp. path-connected, resp. compact, is (X, τ_2) connected, resp. path-connected, resp. compact ? What about the other implication?
8. If $f : X \to Y$ is a continuous map between topological spaces, and X is path connected, is Y path connected?
9. If $f : \mathbb{S}^1 \to \mathbb{R}$ is continuous, does there exist $x \in \mathbb{S}^1$ with $f(x) = f(-x)$?
10. If (X, τ) is *not* simply connected, and $f : X \to Y$ is continuous, can $f(X)$ be simply connected? If instead X is simply connected, is $f(X)$ always simply connected?
11. If X is not simply connected, can $X \times Y$ be simply connected?
12. Let (X, τ) be a topological space and take a subset $A \subseteq X$ with the subspace topology. If A is compact, is $\overline{A}$ (the closure of A) compact? If $\overline{A}$ is compact, is A compact?

*julie.bannwart@uni-mainz.de

2 Session 07/01/2026

These quizzes are completely optional; it is just an additional opportunity to practice and get familiar with the material from the lecture.

1. True or false?
   (a) A compact subspace $A \subseteq X$ cannot be open in X.
   (b) A closed set can have an empty boundary.
   (c) The intersection of two bases (for some topologies) is a basis.
   (d) There exists a topology on $\mathbb{R}$ such that the corresponding topological space is not connected.
   (e) There exists no map from a compact space to a non compact space.
   (f) There exists no map from a non compact space to a compact space.
   (g) Let X have the discrete topology. Every sequence that converges in X is eventually constant.
2. Let X be an infinite set. Is the cofinite topology on X Hausdorff?
3. What is the fundamental group of a torus with one point removed?
4. What is the fundamental group of a Klein bottle (Exercise 7.4) with one point removed?
5. Let $n \geq 1$. What is the fundamental group of $S^n \setminus \{x_1, \ldots, x_k\}$ (the n-sphere with k distinct points removed)?
6. What is the fundamental group of S^2 with the north pole N and south pole S identified, i.e., $S^2/N \sim S$. (Hint: this is equivalent to gluing an interval $[0, 1]$ to S^2 (how?))
7. Let $n \geq 2$. Consider the set of n by n matrices $\mathbb{R}^{n \times n}$ with the Euclidean topology. Which of the following subspaces are compact? (Hint: which theorem about compact subspaces in $\mathbb{R}^m$ can you use?)
   (a) Orthogonal matrices (i.e., $\{M \in \mathbb{R}^{n \times n} \mid M^\top M = I\}$)
   (b) Idempotent matrices (i.e., $\{M \in \mathbb{R}^{n \times n} \mid M^2 = I\}$)
   (c) Diagonalizable matrices
8. *(Harder)* Consider $[0, 1]$ with the Euclidean topology, and take the countable product $\prod_{n \in \mathbb{N}} [0, 1]$ with the product topology. Can you find a metric inducing this topology?

Question 1

True or False:

Let $V \subseteq X$, V connected.

Then $\overline{V}$ is connected.

Question 2

True or False:

Let $U \subseteq X$ dense.

If U is connected, then X is connected.

Question 3

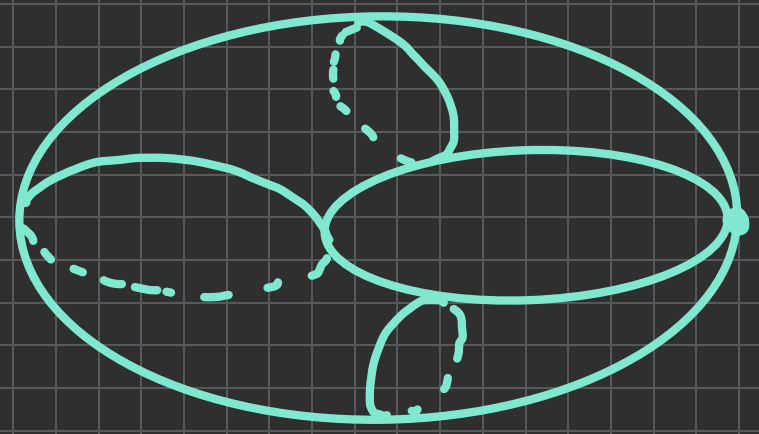
True or False:

Let $U \subseteq X$ dense.

If U is not connected, then X is not connected.

Question 4

Compute π_1 of the "pinched torus"

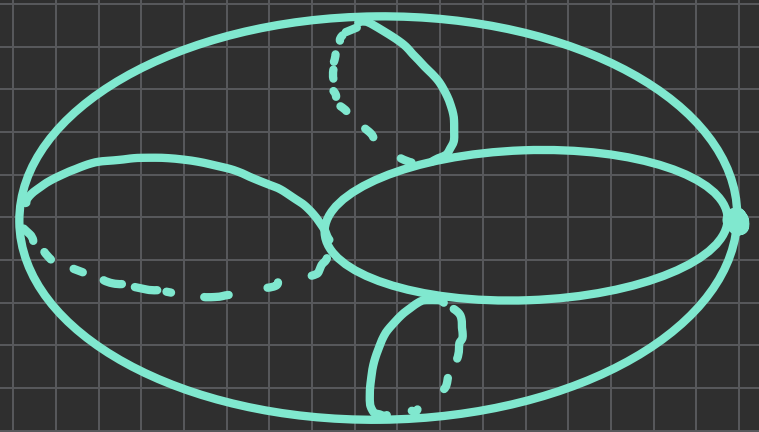


$$S^1 \times S^1 / S^1 \times \{*\}$$

Hint:

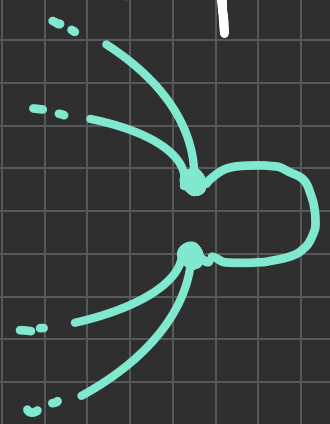
Question 4

Compute π_1 of the "pinched torus"



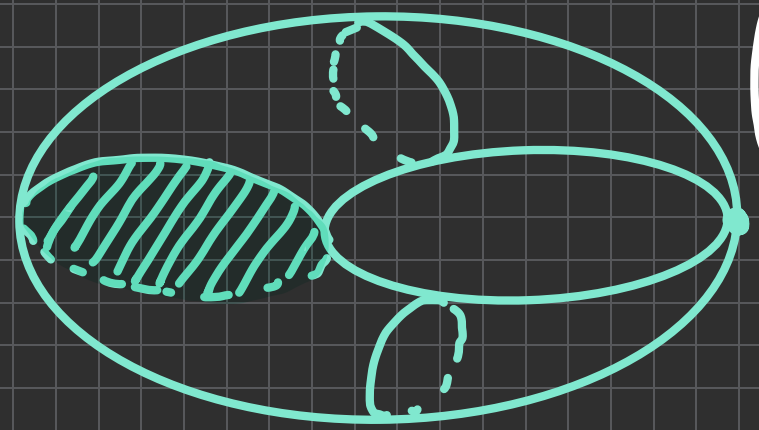
$$S^1 \times S^1 / S^1 \times \{*\}$$

Hint: replace the marked point with an interval



Question 5

Compute π_1 of the "pinched torus" with a disk glued in



$$\left(S^1 \times S^1 / S^1 \times \{*\}\right) \cup_{S^1} D^2$$

Hint: we know how gluing a cell affects π_1 of a space

Question 6

True or False:

Every topological space is a CW-complex

Hint:

Question 6

True or False:

Every topological space is a CW-complex

Hint: what topological properties do CW-complexes have?

Question 7

True or False:

The function $D^2 \longrightarrow S^1 \subseteq \mathbb{R}^2$

$$x \longmapsto \begin{cases} x/\|x\| & \text{if } x \neq 0 \\ (1,0) & \text{else} \end{cases}$$

is continuous.

Question 8

True or False:

Every finite dimensional CW-complex is compact

$\mathcal{X}_{int}$:

Question 8

True or False:

Every finite dimensional CW-complex is compact

Hint: What if we consider only 0-cells?

Question 9

$\mathbb{R}$ can be given the structure of a CW-complex:

- False
- True, but with infinitely many cells only
- True, even with finitely many cells

Question 10

True or False:

If X is a CW complex with n-dimensional cells, then $H_n(X) \neq 0$.

Hint:

Question 10

True or False:

If X is a CW complex with n-dimensional cells, then $H_n(X) \neq 0$.

Hint: simplest example of such a CW complex?

Question 12

A constant simplex $\Delta^n \to X$ is a boundary in the singular complex of X.

- False
- True
- It depends on n.

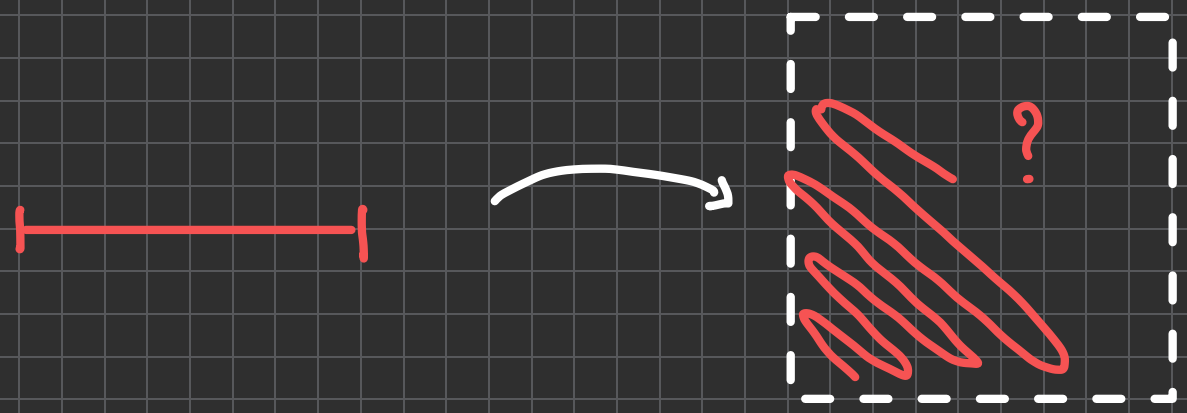
Try low n's!

Question 13

True or False:

There is a continuous surjective map

$$[0,1] \longrightarrow (0,1)^2.$$



Hint:

Question 13

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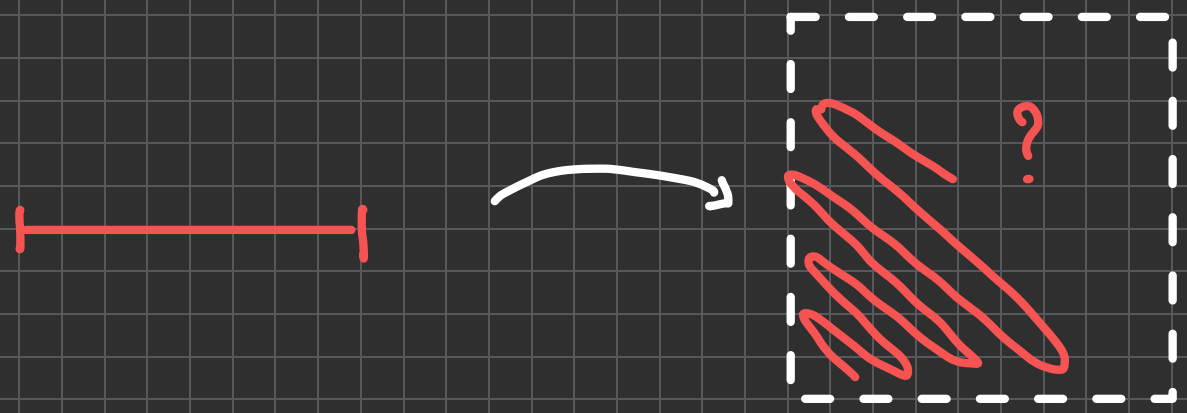


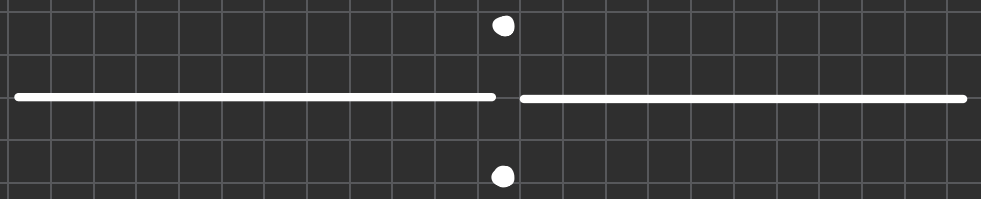
Hint: By Heine-Borel, $[0,1]$ is... ?

Question 14

Consider the line with two origins. Is it:

- connected: Yes - No
- path connected: Yes - No
- Hausdorff: Yes - No

$$\mathbb{R} \times \{0,1\} \Big/ (s,0) \sim (s,1) \ \forall s \neq 0$$



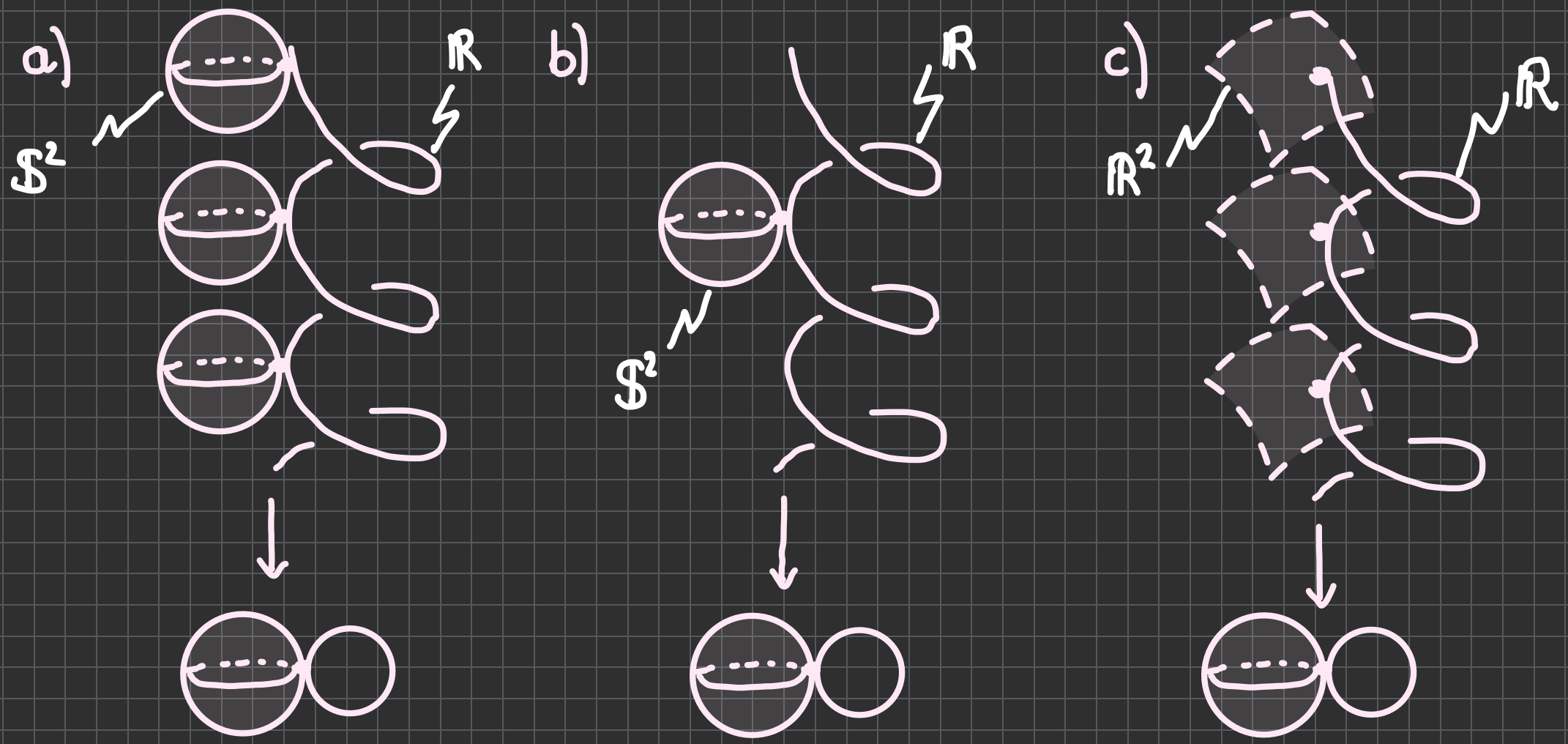
Question 15

Consider $\mathbb{N}$ with the cofinite topology. Is it:

- connected: Yes - No
- compact: Yes - No
- Hausdorff: Yes - No

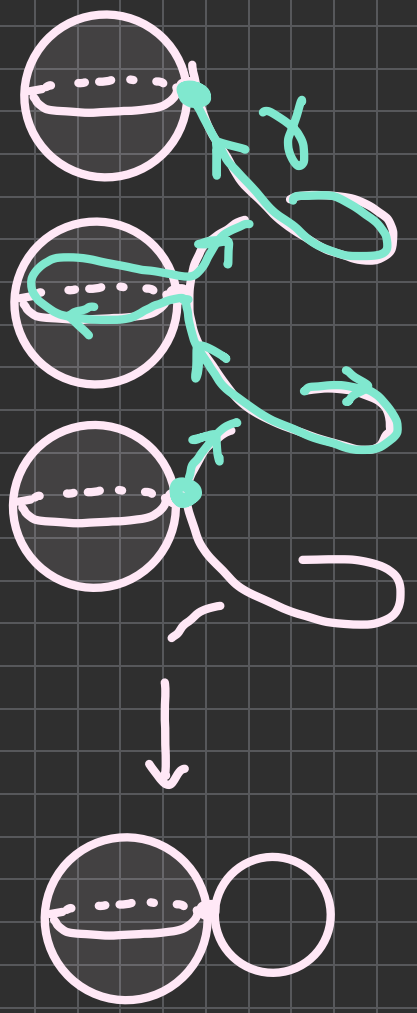
Question 16

What is the fundamental cover of $S^1 \vee S^2$?



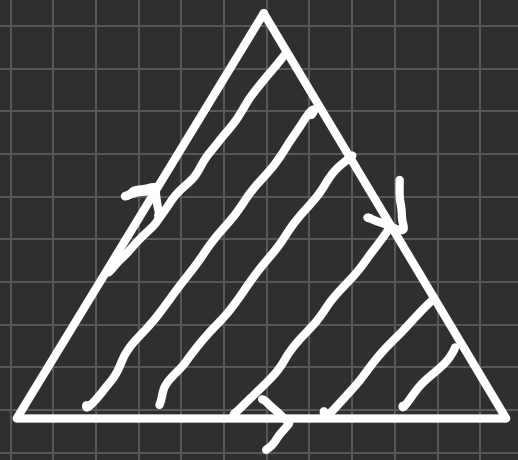
Question 17

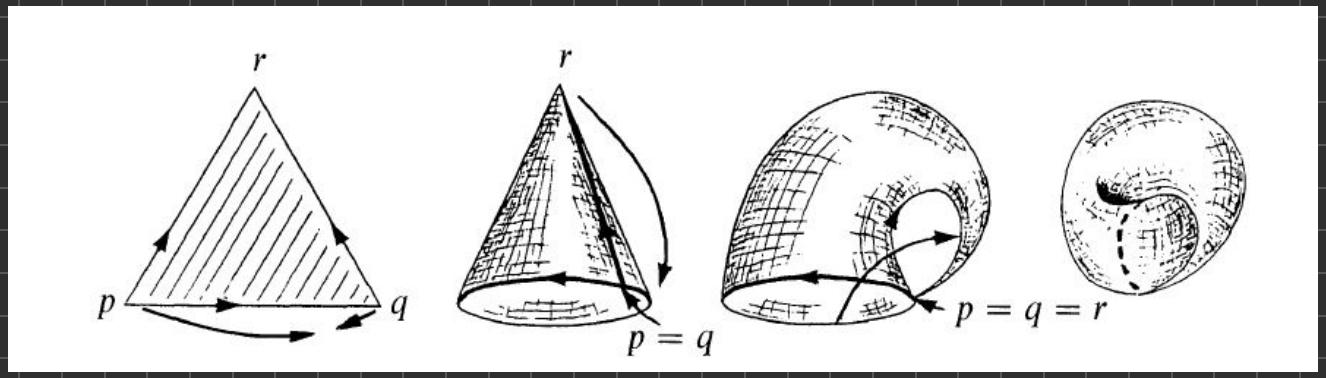
What is $p_*([\gamma])$ in $\pi_1(S^1 \vee S^2) \simeq \mathbb{Z}$?



Question 18

What is the Euler characteristic of the "dunce hat"?

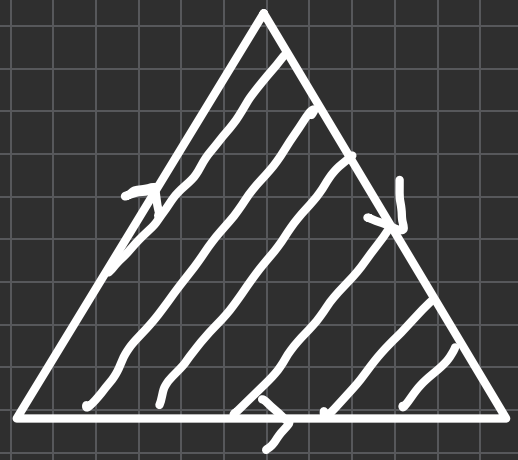


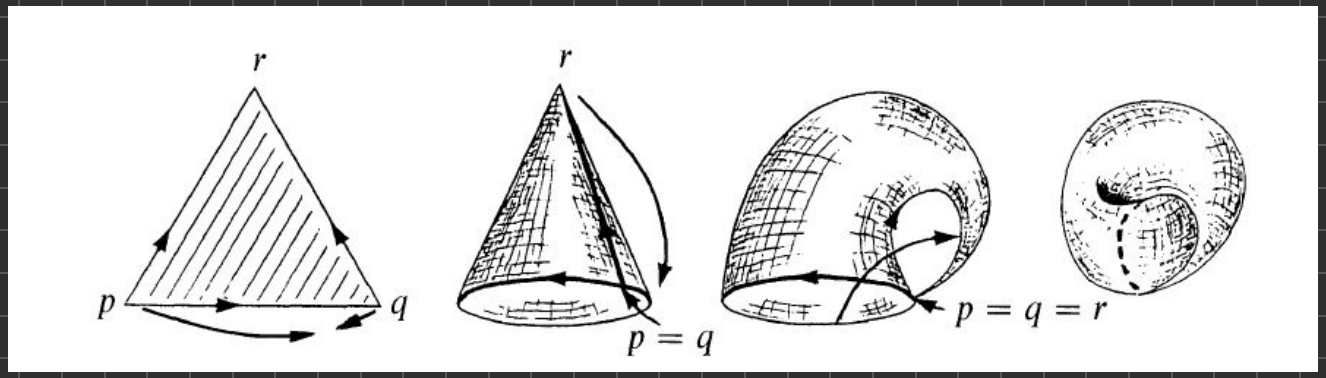


Picture: Topology & Geometry, Glen Bredon

Question 19

What is π_1 of the "dunce hat"?



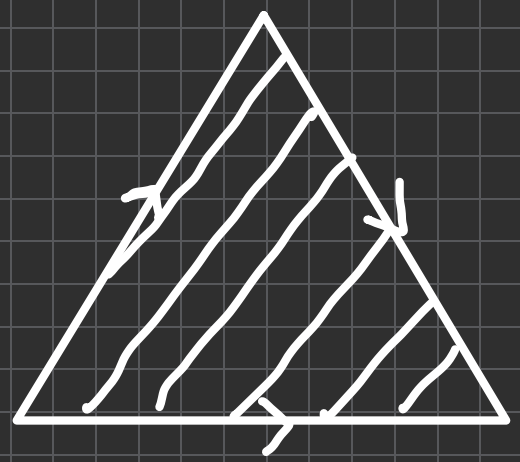


Picture: Topology & Geometry, Glen Bredon

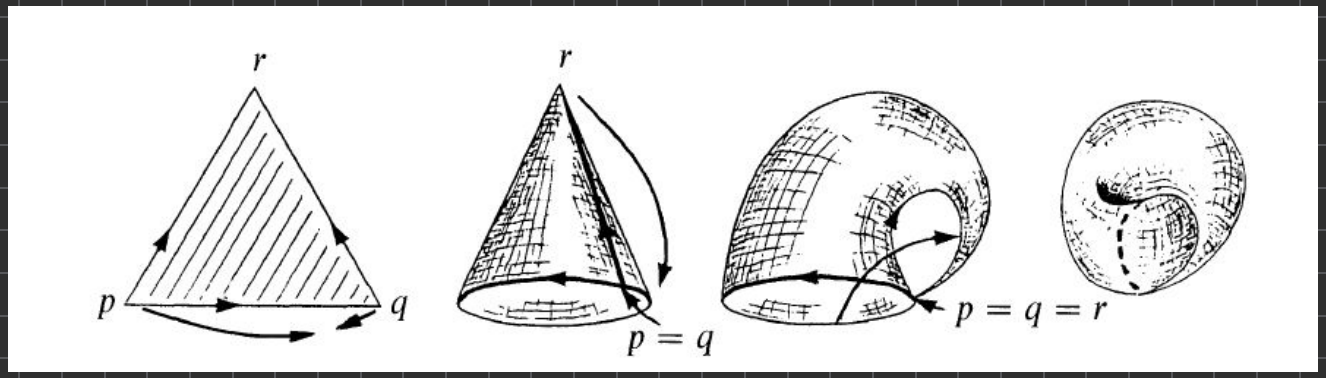
Question 20

Is the "dunce hat" a surface as seen in class ?

Yes - No



Hint: two previous questions + classification of surfaces



Picture: Topology & Geometry, Glen Bredon

Question 21

True or False:

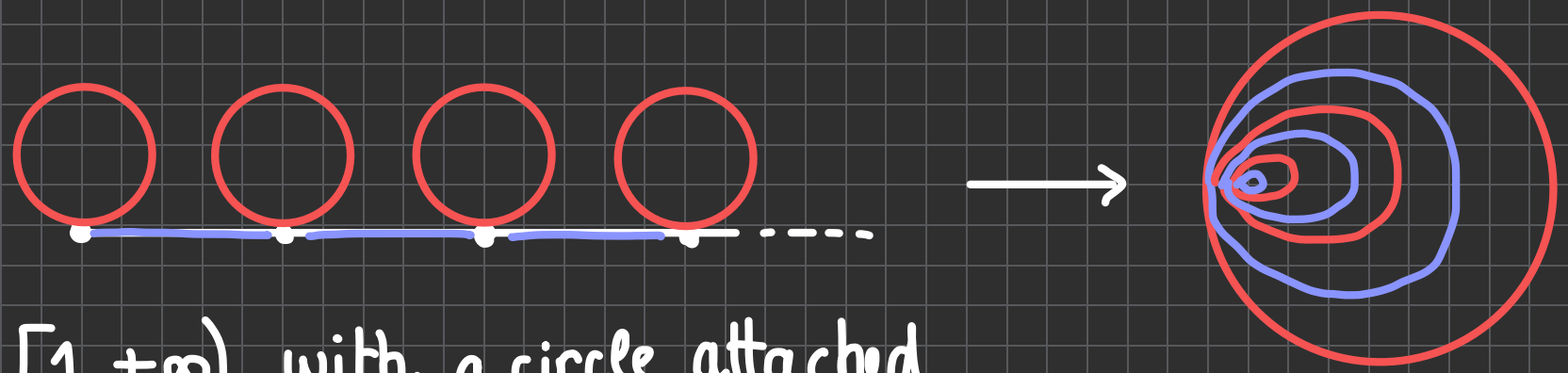
There exists a CW-structure on $\mathbb{S}^2$ with an odd number of 0, 1 and 2-cells.

Hint: a few questions ago...

Question 22

Is this a covering of the Hawaiian earring?

Yes - No



$[1,+\infty)$ with a circle attached to each odd integer

$$\bigvee_{n=1}^{\infty} S^1_n \subseteq \mathbb{R}^2$$

where S^1_n circle of radius $1/n$

Question 23

$[0,+\infty) \vee [0,+\infty) \vee [0,+\infty)$

Does the following define a metric on the tripod?

$$d(x,y) = \begin{cases} \text{Euclidean distance if } x,y \text{ on the same leg} \\ 1 \text{ else} \end{cases}$$

- Yes
- No, and exactly 1 axiom fails
- No, and exactly 2 axioms fail
- No, and exactly 3 axioms fail

Question 24

Compute the homology of $\mathbb{R}^3$ minus three distinct lines through the origin.

Hint:

Question 24

Compute the homology of $\mathbb{R}^3$ minus three distinct lines through the origin.

$\mathcal{X}_{int}: \mathbb{R}^3 \setminus \{0\} \simeq \mathbb{S}^2$